

BHARAT SINGH PARIHAR

AI / Machine Learning Engineer

+91 9451747691 | bharat3645@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Software engineer who is building and shipping deep-learning and LLM systems in production. Trained a CNN deepfake detector to 93.5% accuracy, built a GraphRAG assistant that improved document-QA accuracy close to 4x over baseline RAG, and cut AI inference cost by 58 to 65% on a live product. Comfortable across the stack, from **model training to Go** and FastAPI backends. Author of 1 Q1 Journal springer paper and 3 SCOPUS-indexed IEEE papers and 2 book chapters; CGPA 8.5/10.

TECHNICAL SKILLS

Machine Learning and DL: PyTorch, NumPy, Pandas, Keras, TensorFlow, Scikit-learn, Seaborn, PyTesseract, Unsloth, CNNs, GANs, RNN/LSTM, YOLOv8, OpenCV, NLP, NER, Federated learning, Feature engineering, Hyperparameter tuning

LLM and GenAI: LangChain, LangGraph, RAG, GraphRAG, FAISS, Pinecone, Agentic systems, Prompt engineering, LLM orchestration, Evaluation, Hugging Face

MLOps and Cloud: Docker, Kubernetes, Ollama, AWS (SageMaker, S3, EC2, Lambda), LangFuse, LangSmith, GCP, CI/CD (GitHub Actions), gRPC, W&B, Model deployment and Monitoring, Prometheus, Grafana

Backend and Web: FastAPI, Go, Node.js, REST APIs, Microservices, React, Next.js, Agno, Mastra

Databases: PostgreSQL, Neo4j, MongoDB, Redis, MySQL, Oracle DB

Languages: Python, Go, TypeScript, SQL, C++, Rust

EXPERIENCE

RnR Consulting Pvt. Ltd., Web Developer Intern (AI/ML)

Delhi, India | Dec 2025 - Present

- **Cut AI inference cost by 58%** on large tasks and 65% on small tasks by co-building a **5-module model-routing harness** with token accounting across two enterprise products.
- Built production RAG and agentic inference workflows with SSE streaming and quota-aware request routing across multiple LLM providers.
- Shipped Go microservices that track **data lineage in Neo4j** across a Temporal-orchestrated, 29-service backend serving 500+ concurrent users.

PGDAV College, University of Delhi, Summer Research Intern

May 2024 - Jul 2024

- Built a CNN deepfake and anomaly-detection pipeline to 93.5% accuracy using targeted data augmentation, weighted-layer tuning, and MCDM-weighted ensemble scoring; the methodology fed a SCOPUS-indexed IEEE publication.

KEY PROJECTS

GenAI Realtime Assistant | *LangChain, LangGraph, Neo4j, FAISS*. - Improved document-QA accuracy close to 4x over baseline RAG using GraphRAG retrieval and MMR re-ranking for grounded, diverse answers.

AI-Driven Image Encryption (GAN-LSTM) | *PyTorch, GAN, LSTM, SRDNN*. - Built a 19M-parameter GAN-LSTM model for one-pass image encryption; runs about 50% faster than AES-256 with zero fidelity loss.

Federated Learning Anomaly Detector | *FedML, TFF, gRPC, TLS*. - Reached 91% fraud-detection accuracy across distributed nodes without centralizing client data.

GigX, Decentralized Freelance Marketplace | *Next.js, Solidity, Docker*. - Built smart-contract escrow with on-chain dispute resolution; placed 3rd at BITS Pilani Web 3.0 Hackathon 2025.

Brain Tumor Detection | *YOLOv8, PyTorch, Docker, CUDA*. - Containerized an MRI detector at 89% accuracy with 15% fewer false positives for scalable deployment.

RESEARCH AND ACHIEVEMENTS

- **Publications:** 1 Springer Nature published Q1 Journal paper (2026), 3 SCOPUS-indexed IEEE papers (ICISCT 2024, Kookmin University; ICPCT 2025, Amity) across computer vision, and digital health, plus 2 book chapters (federated learning, renewable-energy AI), 2025.
- **Hackathons:** Winner, GDSC (2023); 3rd, BITS Pilani Web 3.0 (2025); 3rd, IEEE Research (2023); top-5 finishes at Hackronyx and CyberHack (2025).
- **Leadership:** Chair (100+ members) and Founder Vice-Chair, CSI Student Chapter; IEEE Core Member; Lead Organizer, SITNovate 24-Hour Hackathon (200+ participants).

CERTIFICATIONS

AMD GPU Optimization for LLM Inference | AMD Triton Kernel Development | NVIDIA Deep Learning and CUDA C | Stanford Machine Learning Specialization | Google Cloud Data Engineer | AWS Cloud and Cybersecurity Foundations

EDUCATION

Symbiosis Institute of Technology, Nagpur, India

2022 - 2026

B.Tech. (Hons.) Computer Science Engineering, Data Science Specialization | CGPA: 8.5 / 10

Kendriya Vidyalaya, Unnau | HSC (PCM with Computer Science), 87.44%, 2022